Unit 5 Pre-Test

The Pythagorean Theorem

Standards assessed: NC.8.G.6, NC.8.G.7, NC.8.G.8

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.G.7 Apply the Pythagorean theorem to find the hypotenuse.

• *I can use $a^2 + b^2 = c^2$ to find the hypotenuse of a right triangle.*

Match each vocabulary term to its definition, then find each hypotenuse.

C hypotenuse ✓	**A.** The two sides that form the right angle.
A legs ✓	**B.** $a^2 + b^2 = c^2$, relating the sides of a right triangle.
B Pythagorean theorem ✓	**C.** The side opposite the right angle; the longest side.
D right triangle ✓	**D.** A triangle that contains one 90° angle.
E square root ✓	**E.** A number that, multiplied by itself, gives a given value.
F converse ✓	**F.** If $a^2 + b^2 = c^2$, then the triangle is a right triangle.

Pythagorean theorem — reference

$$a^2 + b^2 = c^2$$

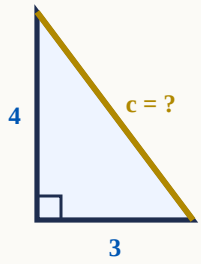
c *is the hypotenuse (opposite the right angle);* **a** *and* **b** *are the legs.*

1. *Find c*

Legs 3 and 4. Find the hypotenuse.

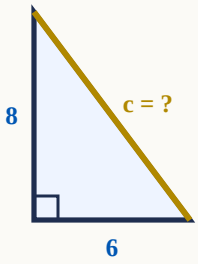
2. *Find c*

Legs 6 and 8. Find the hypotenuse.



c =

answer: 5

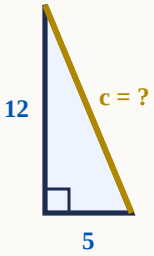


c =

answer: 10

3. *Find c*

Legs 5 and 12. Find the hypotenuse.

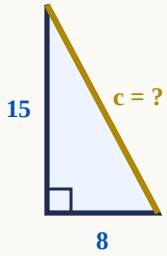


c =

answer: 13

4. *Find c*

Legs 8 and 15. Find the hypotenuse.

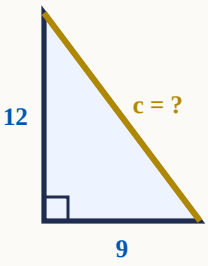


c =

answer: 17

5. *Find c*

Legs 9 and 12. Find the hypotenuse.

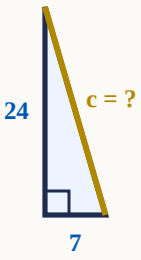


c =

answer: 15

6. *Find c*

Legs 7 and 24. Find the hypotenuse.



c =

answer: 25

SCORE *4.0* **Target Content**

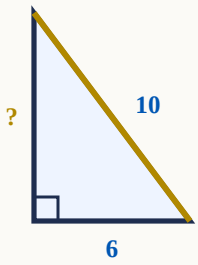
NC.8.G.7 Find a missing leg and estimate non-perfect square roots.

• *I can find a missing leg and round answers to the nearest tenth when needed.*

Find each missing side. Round to the nearest tenth where asked.

7. *Find a leg*

One leg 6, hypotenuse 10. Find the other leg.



a = _____

answer: 8

8. *Find a leg*

One leg 5, hypotenuse 13. Find the other leg.

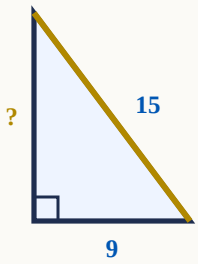


a = _____

answer: 12

9. *Find a leg*

One leg 9, hypotenuse 15. Find the other leg.

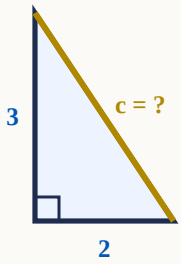


a = _____

answer: 12

10. *Nearest tenth*

Legs 2 and 3. Find the hypotenuse to the nearest tenth.



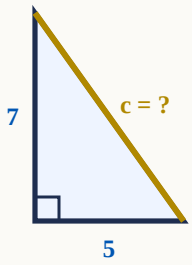
c ≈ _____

answer: 3.6

11. *Nearest tenth*

12. *Nearest tenth*

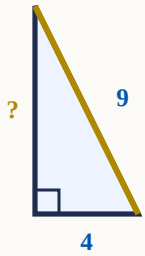
Legs 5 and 7. Find the hypotenuse to the nearest tenth.



c ≈

answer: 8.6

One leg 4, hypotenuse 9. Find the other leg to the nearest tenth.



a ≈

answer: 8.1

SCORE 5.0 Advanced Content

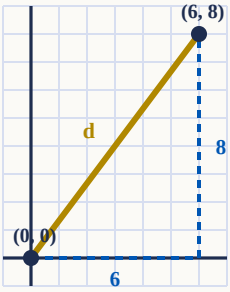
NC.8.G.8 Apply the theorem to the coordinate plane and real-world problems.

• *I can find the distance between two points and solve real-world right-triangle problems.*

Solve each problem.

13. *Distance*

Find the distance between (0, 0) and (6, 8).

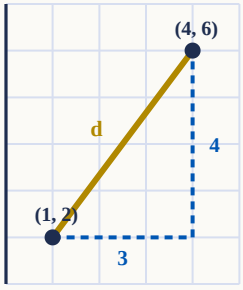


d =

answer: 10

14. *Distance*

Find the distance between (1, 2) and (4, 6).



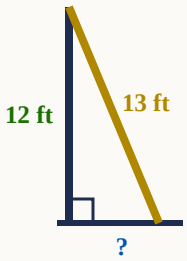
d =

answer: 5

15. *Apply*

16. *Apply*

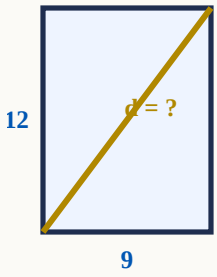
A 13 ft ladder reaches 12 ft up a wall. How far is the base from the wall?



distance = _____

answer: 5 ft

A rectangle is 9 by 12. Find the length of its diagonal.

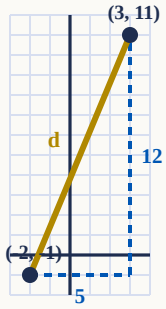


d = _____

answer: 15

17. *Distance*

Find the distance between (−2, −1) and (3, 11).

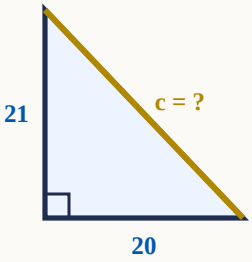


d = _____

answer: 13

18. *Apply*

A park is a right triangle with legs 20 m and 21 m. Find the longest side.



longest side = _____

answer: 29 m